THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A man stands 3 m in front of a plane mirror and a boy stands 2 m behind the man. The distance of the boy's image from the man is: 1

(a) 5 m **(b)** 3 m **(c)** 8 m **(d)** 2 m

2. A total lunar eclipse is seen at midnight from a place. At that time the Moon is: 1

(a) close to the Sun **(b)** below the horizon

(c) high in the sky **(d)** not present in the sky

3. If a pesticide kills all the frogs in a field, the number of insects will most likely: 1

(a) increase **(b)** decrease **(c)** become constant **(d)** stay zero

4. A student says the Earth's inner core is liquid because it is hotter than the outer core. This reasoning is wrong because: 1

(a) pressure raises the melting point **(b)** iron cannot be solid

(c) the inner core is gas **(d)** the outer core is cooler

5. A student heats three metal rods of equal size and finds wax dropping at different times. The dependent variable is: 1

(a) time for wax to drop **(b)** flame

(c) metal type **(d)** length of rod

6. Which is NOT a feature of a unicellular organism? 1

(a) nucleus (in protozoa) **(b)** cytoplasm

(c) tissues **(d)** cell membrane

7. A new disease spreads from person to person. Which first step reduces the spread? 1

(a) closing hospitals **(b)** isolating the patients and hygiene

(c) stopping all food **(d)** ignoring symptoms

8. A student's 'magnet' was a steel rod in a coil and kept its magnetism after the current stopped. This shows that steel: 1

(a) retains magnetism **(b)** loses magnetism quickly

(c) is an insulator **(d)** cannot be magnetised

9. To reduce the wear of the surfaces of the moving parts of a machine, we should: 1

(a) increase the load **(b)** apply a lubricant

(c) use a rough material **(d)** stop the machine

10. A coastal village plants mangroves to: 1

(a) create wind **(b)** stop wind completely

(c) reduce the force of the surge **(d)** increase rain

11. Which has the weakest forces between its particles at room temperature? 1

(a) ice **(b)** oxygen **(c)** iron **(d)** water

12. A shiny, hard element conducts electricity and turns black in air with sulphur compounds. It could be: 1

(a) sulphur **(b)** carbon **(c)** neon **(d)** silver

13. A sample of tap water leaves white deposits on evaporation. This shows that it contains: 1

(a) oil **(b)** dissolved solids

(c) pure water **(d)** dissolved gases only

14. A convex mirror is fitted in shops to see a wide area because it forms: 1

(a) an inverted image of the same size

(b) no image

(c) an erect, diminished image that covers a wide field

(d) an enlarged real image

15. After 33 years the lunar and solar years differ by about: 1

(a) one year **(b)** one month **(c)** ten years **(d)** one day

16. Lichen is an association of: 1

(a) algae and animals **(b)** bacteria and plants

(c) algae and fungi **(d)** fungi and animals

17. A town draws more water from wells than rain recharges. In the long term: 1

(a) ozone increases **(b)** the water table rises

(c) wells overflow **(d)** the water table falls

18. Assertion (A): Records should include observations that do not match the expectation. Reason (R): Selective reporting gives an incomplete and misleading picture. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

19. Assertion (A): A red blood cell can carry more oxygen because it has no nucleus. Reason (R): Haemoglobin takes up the space gained. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. Assertion (A): Diabetes is a communicable disease. Reason (R): Diabetes is not caused by any pathogen. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

21. Assertion (A): A compass needle placed under a current-carrying wire deflects. Reason (R): Electric current produces a magnetic field around the wire. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

22. Assertion (A): The weight of an object is less on the Moon than on the Earth. Reason (R): The Moon has weaker gravity than the Earth. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

23. Assertion (A): The pressure exerted by a person is more when standing on one foot. Reason (R): The area of contact is half. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

24. Assertion (A): A steam burn is more severe than a burn from boiling water. Reason (R): Steam gives out extra heat on condensing. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

25. Assertion (A): Graphite conducts electricity although carbon is a non-metal. Reason (R): Graphite is an exception among non-metals. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

26. Assertion (A): The mass of a solution is equal to the sum of the masses of solute and solvent. Reason (R): Mass is conserved during dissolving. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

27. Assertion (A): A convex lens can start a fire using sunlight. Reason (R): It converges parallel rays to a point. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

28. Assertion (A): The Moon is visible at night. Reason (R): The Moon produces its own light. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

29. Assertion (A): Decomposers are essential for the recycling of nutrients. Reason (R): They convert dead organic matter into simple inorganic substances. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

30. Assertion (A): The Moon has large changes in surface temperature between day and night. Reason (R): The Moon has a thick atmosphere. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

31. A pendulum completes 30 swings in 48 s. The time period is: 1

(a) 78 s **(b)** 0.625 s **(c)** 16 s **(d)** 1.6 s

32. Which sequence correctly lists in increasing size: red blood cell (7 micrometres), bacterium (2 micrometres), virus (0.1 micrometre)? 1

(a) red blood cell, bacterium, virus **(b)** virus, red blood cell, bacterium
(c) virus, bacterium, red blood cell **(d)** bacterium, virus, red blood cell

33. A sick person drinks water from a well that is next to a leaking toilet pit. The disease most likely to spread is: 1

(a) typhoid **(b)** rickets **(c)** goitre **(d)** scurvy

34. A device connected to a 5 A fuse draws 3 A and a second device draws 4 A. If both are on in the same circuit, the fuse will: 1

(a) melt **(b)** become magnetic **(c)** remain intact **(d)** cool

35. When two forces F1 = 5 N (right) and F2 = 8 N (left) act on a body, a third force of what size and direction makes the net force zero? 1

(a) 13 N to the left **(b)** 3 N to the left
(c) 3 N to the right **(d)** 13 N to the right

36. A man of weight 800 N exerts 40000 Pa when wearing shoes. The area of contact of both shoes is: 1

(a) 0.2 square metre **(b)** 2 square metres
(c) 32 square metres **(d)** 0.02 square metre

37. Which of these is the best explanation for the cooling in a desert cooler? 1

(a) pads produce cold
(b) fan makes cold
(c) water is cold
(d) water evaporates and takes heat from the air

38. Which of the following represents 3 molecules of oxygen? 1

(a) $3O_2$ **(b)** $6O$ **(c)** O_6 **(d)** $3\,O$

39. How much water must be added to 100 g of a 40 per cent solution to dilute it to 10 per cent? 1

(a) 300 g **(b)** 100 g **(c)** 30 g **(d)** 400 g

40. A clock's image in a plane mirror shows 10:10 in a mirror. The actual time is: 1

(a) 11:50 **(b)** 10:10 **(c)** 2:50 **(d)** 1:50

41. **Assertion (A):** Metals are generally good conductors of heat and electricity. 1
Reason (R): Metals are malleable and ductile.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** A soft drink fizzes when its cap is opened. 1
Reason (R): Carbon dioxide dissolved under pressure escapes when the pressure is reduced.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** Convex mirrors are used as rear-view mirrors in vehicles. 1
Reason (R): A convex mirror gives a wider field of view.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** We always see the same side of the Moon from the Earth. 1
Reason (R): The Moon takes the same time to rotate once on its axis as to revolve once around the Earth.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** Decomposers are essential in an ecosystem. 1
Reason (R): Decomposers break down dead plants and animals and return nutrients to the soil.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** Weather events such as rain occur mostly in the troposphere. 1
Reason (R): The troposphere contains most of the air and water vapour of the atmosphere.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** A single measurement is enough to be sure of a result. 1
Reason (R): Repeating a measurement and taking the average reduces the effect of random errors.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** Animal cells have chloroplasts. 1
Reason (R): Chloroplasts contain chlorophyll and carry out photosynthesis.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** Diabetes is a communicable disease. 1
Reason (R): Communicable diseases are caused by germs and can spread from person to person.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** A fuse wire is made of a material with a high melting point. 1
Reason (R): A fuse protects a circuit by melting and breaking it when the current becomes too large.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**

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